

9(a)(i)	$9.0 / \sqrt{2} =$ 6.4 V	A1
9(a)(ii)	$\omega = 20$ $\omega = 2\pi / T$ $T = 2\pi / 20$	C1
	$T = 0.31$ s	A1
9(b)	the r.m.s. voltages are different, so no	B1
9(c)(i)	$P = V_{\text{r.m.s.}} \times I_{\text{r.m.s.}}$	C1
	$= 120 \times 0.64$ $= 76.8$ W	C1
	efficiency $= (76.8 / 80) \times 100$ $=$ 0.96 or 96 %	A1
9(c)(ii)	Any one from: • heat losses due to resistance of windings / coils • heat losses in magnetising and demagnetising core / hysteresis losses in core • heat losses due to eddy currents in (iron) core • loss of flux linkage	B1